HACKATHON PROJECT OVERVIEW

Project Name: Urban-Eye

Problem Statement & ID: AI-Powered Autonomous Drone Response for Urban Safety & 04

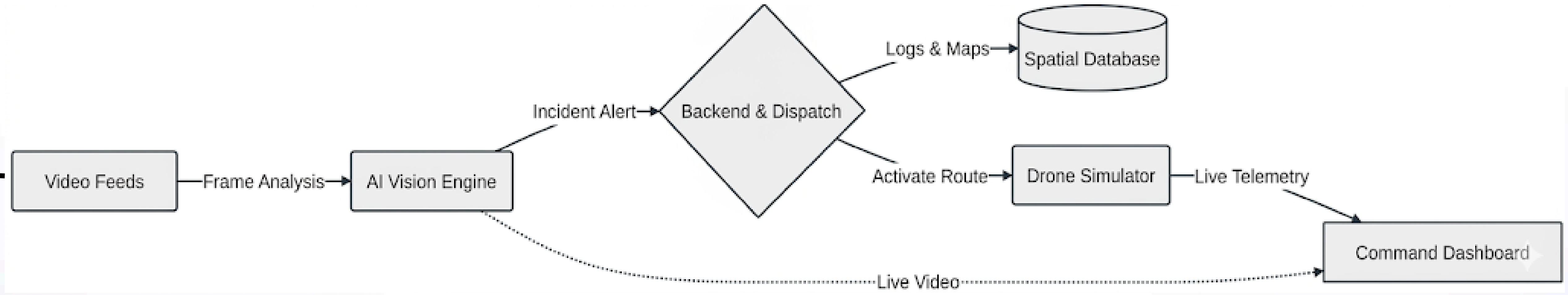
Team Name: Orbital Orbs

PROPOSED SOLUTION

- **Solution Overview:**
  We are building a proactive, **AI-driven simulation platform designed to enhance urban safety**. The system actively monitors **video feeds to detect emergencies** automatically, associates them with geographical coordinates, and **dispatches the nearest simulated autonomous drone** to provide real-time situational awareness to a centralized control center.

- **How the solution works:**
  The system operates on a seamless, automated pipeline:

- **Key components of the solution**
  - **Computer Vision Engine**: Analyzes live or simulated video feeds to detect anomalies.
  - **Simulated Dispatch & Telemetry Service**: Generates coordinate paths and ETA for the dispatched drones.
  - **Command & Control Dashboard:** A unified web interface displaying live map tracking, active alerts, and drone status.
- **Cutting-edge technologies:**
  - Integration of **Edge AI for low-latency incident detection**.
  - **Predictive high-risk zone heatmaps to pre-position drones.**
  - Intelligent alert prioritization to filter out false positives.

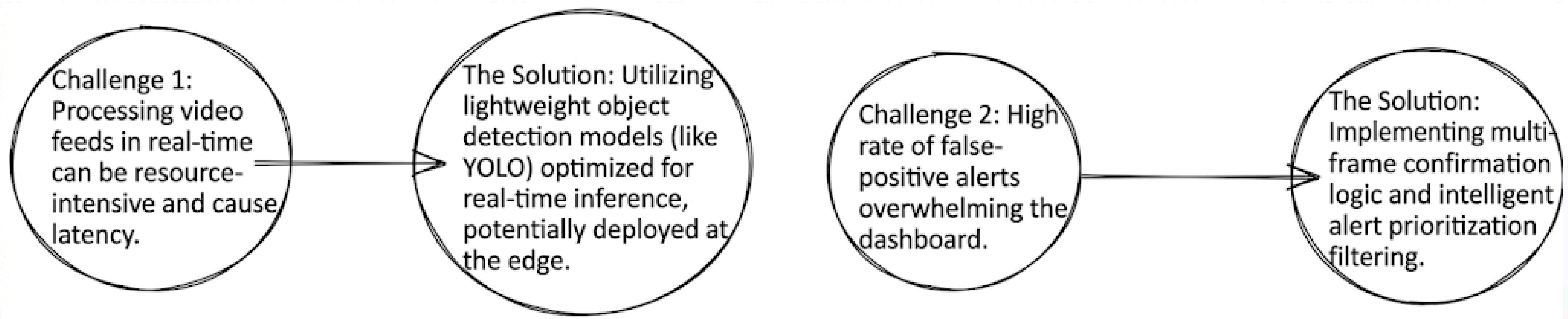


APPROACH

Step-by-step explanation of the approach & Logical breakdown of the process:

- **<u>Ingest & Detect</u>**: **YOLOv8s** runs real-time inference across 200+ GPS-mapped cameras in Pune to detect crowds, fires, and collisions.
- **<u>Score & Filter</u>**: A custom **Severity Engine scores threats 1–10**. Scores **< 5 are manual checks** (in case of false alarms); **≥ 5 trigger instant auto-dispatch**.
- **<u>Navigate & Dispatch</u>**: The system uses the **Haversine formula** to find the closest drone and a pathfinding to route it safely around no-fly zones.
- **<u>Alert & Overwatch:</u> MSG91** instantly texts a live feed link to agencies. The drone hovers at 100m and auto-returns (RTH) on low battery.
- **<u>Human-in-the-Loop</u>**: **UP/DOWN flags on alerts** in the online drone feed from the authorities or operators are logged as ground-truth labels for continuous AI model retraining.

Key Challenges & How They Are Addressed:



USP and Features

Simplify Healthcare

Unique Selling Points (USPs):

What makes the solution special?
Unlike generic alert systems that just flag incidents, **Urban-Eye explains why a threat is CRITICAL factoring confidence, zone risk, object proximity, and historical patterns, so operators act on reasoning, not instinct**. Every **UP/DOWN SMS reply from authorities** becomes a ground truth training signal, continuously recalibrating the model. **The system doesn't just respond, it learns.**

How does it outperform alternatives?
While alternatives rely on manual human dispatch, our architecture automatically handles multi-incident routing. It dynamically **calculates the fastest ETA and navigates around geofenced no-fly zones to deliver live**, on-scene intelligence before ground teams even move.

How does it overcome drawbacks in other solutions?
Existing systems detect but don't dispatch, alert but don't prioritize, and generate false positives that responders learn to ignore. Urban-Eye overcomes this by combining **real-time severity scoring with autonomous drone dispatch** and an authority feedback loop that filters noise and improves accuracy over time.

Key Features:

Incident Detection
Automatically detects crowds, accidents, suspicious vehicles, and abandoned objects.

Smart Navigation
Includes route optimization and no-fly zone avoidance.

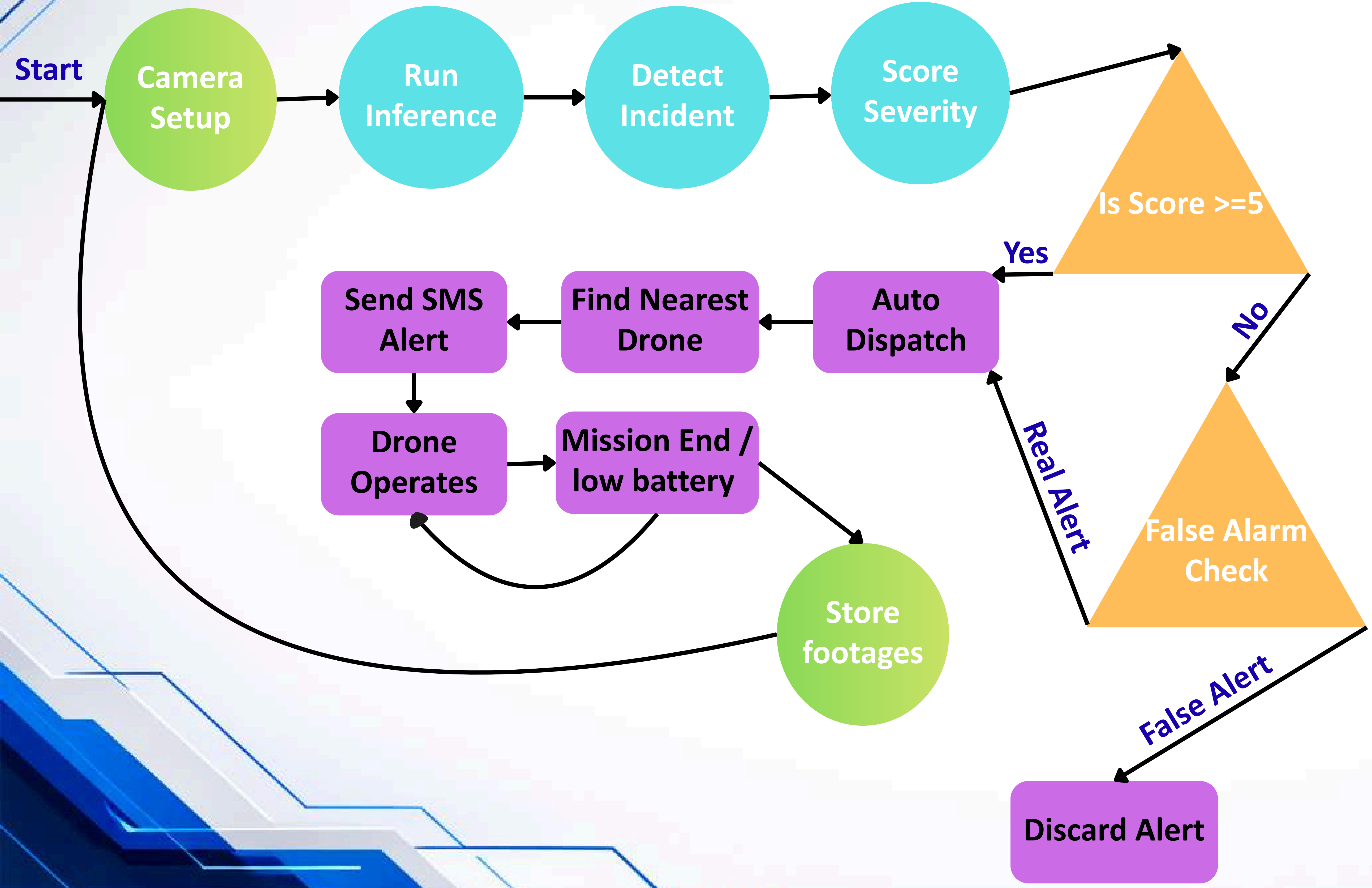
Drone Visualization
Displays active drone routes, ETA, and position updates on an interactive map.

Unified Dashboard
Displays live locations, active alerts with severity indicators, and simulated drone telemetry.

TECHNOLOGIES USED AND IMPLEMENTATION

Simplify Healthcare

Implementation Flowchart:



AI/ML:
YOLOv8s (Ultralytics) - pretrained on **COCO dataset**, best balance of speed vs accuracy for real-time inference.

Programming Languages:

Frameworks:

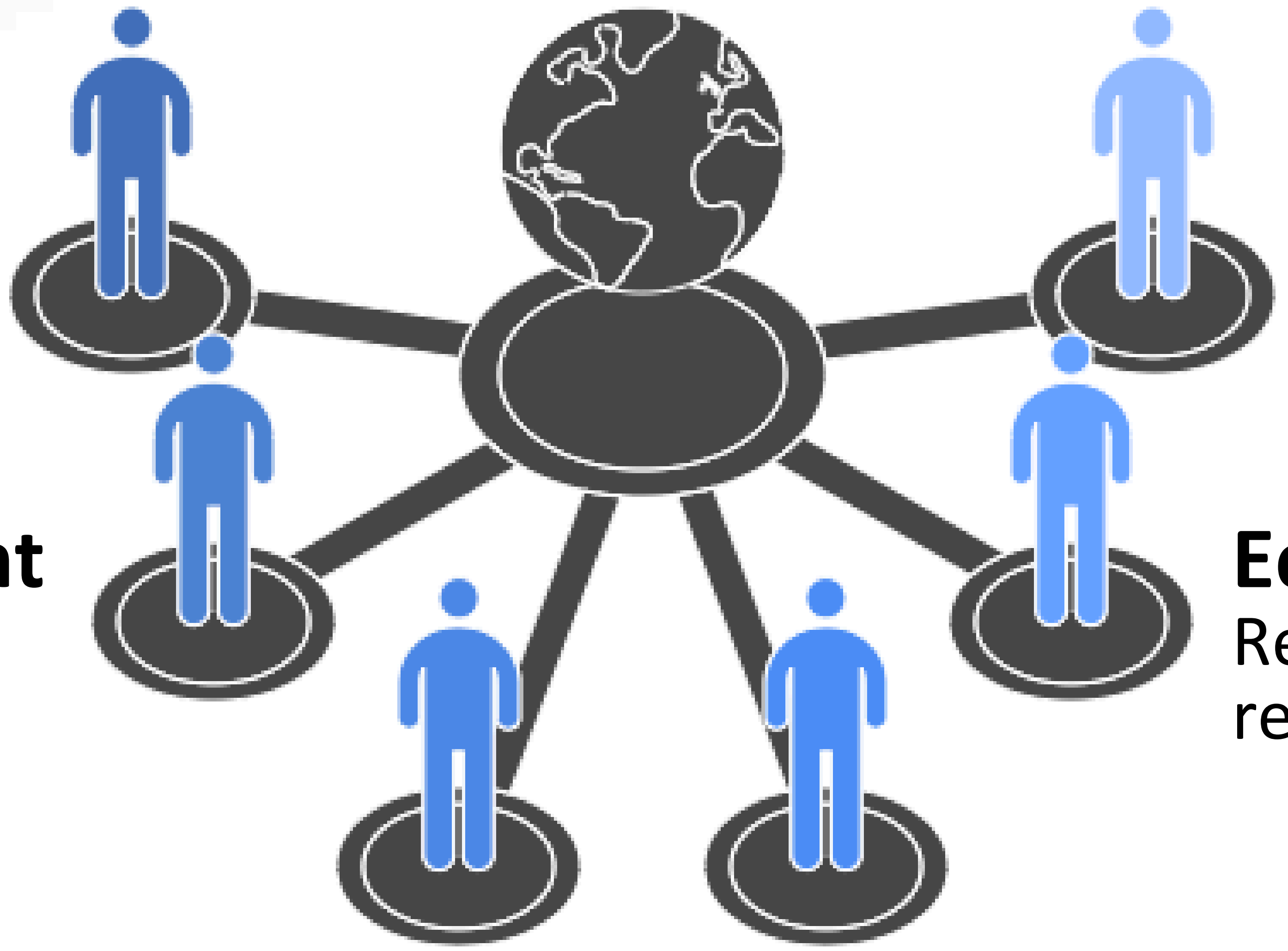
Flask , Ultralytics (YOLO inference)

Databases:
alerts.json flat file (prototype) → TimescaleDB + PostGIS + Redis (production)

Frontend:
Leaflet.js, leaflet.heat plugin, OpenStreetMap tiles

Text:
MSG91 SMS API

POTENTIAL IMPACT



Future Scalability
Integration with real drones and expanded applications.

Enhanced Public Safety
Faster emergency responses save lives.

Improved Law Enforcement
Increased efficiency and digital evidence.

Economic Optimization
Reduced costs through efficient resource deployment.

Technological Advancement
Promotes AI, IoT, and smart city infrastructure.

Environmental Benefits
Lower energy consumption compared to ground vehicles.

REFERENCES & LINKS

- Pretrained model: YOLOv85 - https://github.com/ultralytics/ultralytics
- Map library: Leaflet.js — https://leafletjs.com
- Heatmap plugin: leaflet.heat — https://github.com/Leaflet/Leaflet.heat
- SMS API: MSG91 - : https://docs.msg91.com/reference/send-sms
- Dataset: COCO (Common Objects in Context) — https://cocodataset.org
- UAVs in Search and Rescue:
  https://www.sciencedirect.com/science/article/abs/pii/S2212420925000238

Prototype Link

https://github.com/Nickseth02/Autonomus-drone-response.git

TEAM DETAILS

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